

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0844

Campground 100 kV
December 21, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0844. The Interconnection Customer proposes a 58.4 MW solar photovoltaic facility interconnecting at the Campground 100 kV substation in the Illinois area. The request was received on August 03, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 328 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,537,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Trego-Highgate 230kV	Trego-Troy Energy Llc 500kV	112.4	395	10.9
Farmersville-Kendallville 100kV L	Dekalb Substation-Jones Chapel 500kV	111.9	205	5.4
Richland 500/230 kV TX	Troy Energy Llc-Richland 500kV L	111.6	1,398	32.8
Kendallville-Rex Brown 100kV	Dekalb-Jones Chapel 500kV	111.0	216	5.4
Rex Brown-Montibello 100kV	Jones Chapel-Dekalb 500kV line	109.9	233	5.4
Highgate-Troy Energy Llc 230kV	Troy Energy Llc-Trego 500kV	109.5	755	12.4
Ligonier-Montibello Substation 100kV	Dekalb-Jones Chapel 500kV	109.2	244	5.4
Lookingglass-Corridor Substation 230kV	Beaver-Trego 500kV	109.0	608	9.9
Jolie 29-43Rd St Terminal 230kV Line	Pueblo Xtra-Boca Raton 500kV	108.5	816	5.6
Twin County Ranch-Richland 230kV	Troy Energy Llc-Richland 500kV	107.4	822	19.5
Cuba City-Farmersville 100kV	Dekalb-Jones Chapel Substation 500kV	106.2	197	5.4
Minnehaha-Sidney A Murray Jr Hydroelectric 230kV line	Dekalb-Richland 500kV line	104.1	844	16.6
N Branch Mn-Cayuga 100kV #2	North Branch Mn-Cayuga 100kV Line 1	103.0	146	8.6
Lookingglass-Truman 230kV	Trego-Beavr 500kV	102.7	673	9.9
Cayuga-North Branch Mn 100kV #1	N Branch Mn-Cayuga 100kV Line	102.6	147	8.6
George Neal South-Myrtle Point 230kV	Galion Generating Station-Trego 500kV	102.5	670	6.5
Zona Libre 6007-George Neal S 230kV	Galion Generating Sta-Trego 500kV	102.2	694	6.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Truman-Waldo 230kV Line	Beaver-Trego Substation 500kV	102.1	700	9.9
Zona Libre 6007-Kampsville 230kV	Galion Generating Sta-Trego 500kV L	102.0	719	6.5
Moreno Tap-Kampsville 230kV	Trego-Galion Generating Station 500kV Line	101.9	733	6.5
Eastown Road-Moreno Tap 230kV	Galion Generating Sta-Trego 500kV	101.6	758	6.5
Trego-Troy Energy Llc 500kV	Trego-Swan Vly 500kV Line	101.6	2,800	32.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Trego-Highgate 230kV	Rebuild line with higher-capacity conductor	\$36,531,000	0% (\$191,000)
Farmersville-Kendallville 100kV line	Rebuild line with higher-capacity conductor	\$5,062,000	0% (\$26,000)
Richland 500/230 kV TX	Replace transformer with higher-rated unit	\$25,356,000	0% (\$133,000)
Kendallville-Rex Brown 100kV	Rebuild line with higher-capacity conductor	\$3,014,000	0% (\$16,000)
Rex Brown-Montibello 100kV	Reconductor line	\$2,322,000	0% (\$12,000)
Highgate-Troy Energy Llc 230kV	Reconductor line	\$15,386,000	0% (\$80,000)
Ligonier-Montibello Substation 100kV	Reconductor line	\$2,315,000	0% (\$12,000)
Corridor-Lookingglass 230kV	Reconductor line	\$7,269,000	0% (\$38,000)
43Rd St Terminal-Joliet 29 230kV	Reconductor line	\$2,109,000	0% (\$8,000)
Twin County Ranch-Richland 230kV	Reconductor line	\$17,652,000	0% (\$93,000)
Cuba City-Farmersville 100kV	Reconductor line	\$1,872,000	0% (\$10,000)
Minnehaha-Sidney A Murray Jr Hydroelectric 230kV	Reconductor line	\$14,918,000	0% (\$78,000)
N Branch Mn-Cayuga 100kV #2	Reconductor line	\$434,000	100% (\$434,000)
Truman-Lookingglass 230kV	Reconductor line	\$5,251,000	0% (\$27,000)
Cayuga-North Branch Mn 100kV #1	Reconductor line	\$665,000	100% (\$665,000)
George Neal South-Myrtle Point 230kV	Reconductor line	\$6,357,000	0% (\$33,000)
Zona Libre 6007-George Neal S 230kV	Reconductor line	\$11,620,000	0% (\$60,000)
Truman-Waldo Substation 230kV	Reconductor line	\$6,683,000	0% (\$35,000)
Zona Libre 6007-Kampsville 230kV	Reconductor line	\$9,511,000	0% (\$49,000)
Kampsville-Moreno Tap 230kV	Reconductor line	\$12,748,000	0% (\$66,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Eastown Rd-Moreno Tap 230kV line	Reconductor line	\$23,466,000	0% (\$121,000)
Trego-Troy Energy Llc 500kV	Reconductor line	\$67,393,000	0% (\$350,000)
	Total allocated to project		\$2,537,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.